MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 8945

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Coaxial tunable frequency, 8.7 to 9.4 GHz, integral magnet, pulsed, 220 kw peak power output, air cooled.

ABSOLUTE RATINGS:

Parameter:	If(surge)	If	tk	epy	VSWR	Ef	ib	Pi	pi	Input Alt.
Unit:	a	A	sec	kv	----	V	A	W	kw	ft.
Maximum:	12	3.3	---	24	1.5:1	15	30	690	690	70,000
Minimum:	--	2.9	150	---	---	Note 5	15	---	---	(Note 4)

ABSOLUTE RATINGS:

Parameter:	Du	tpc	rrv	Tbody	Tuner Drive Torque	Pressurization Input	Pressurization Output
Unit:	----	μs	kv/μs	°C	in-oz	psia	psia
Maximum:	.0013	2.75	160	150	200	45	45
Minimum:	-----	0.2	70	-55	---	11	11
		Note 6	Note 7	Note 2	Note 3	Note 4	

PHYSICAL CHARACTERISTICS:

Dimensions:	Figure 1	Mounting Position:	Any
Weight:	13 lbs. approx.	Marking:	See note 32
Coupling:	See note 25	Cooling:	Forced air See note 10

TEST CONDITIONS:

Test 1: (See notes 26 and 28)

Parameter:	tk	Ef	Ib	Du	tpc	rrv	Tbody
Unit:	sec	V	mA	--	μs	kv/μs	°C
Maximum:	150	13.75	--	--	2.5	180	100
Minimum:	---	---	--	.001	1.9	160	80
Nominal:	---	---	27.5	--	2.3	---	---
Notes:		5					

Test 2:

Parameter:	tk	Ef	Ib	Du	tpc	rrv	Tbody
Unit:	sec	v	mA	--	μs	kv/μs	°C
Maximum:	---	5.0	---	--	0.5	180	100
Minimum:	150	---	---	.0008	0.3	160	80
Nominal:	---	---	19.2	--	0.4	170	---
Notes:		5					

GENERAL:

Preproduction Test - Required (See note 44)

Frequency

F1 = 8700 ± 5 MHz
F2 = 9000 ± 5 MHz
F3 = 9400 ± 5 MHz

8945

FSC 5960

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 Notes 1, 2, 6 and 28						
4003	Pressurization	-	45 minimum psia input & output assemblies. See notes 22 and 23	---	---	---	---
1301	Heater current	-	Ef = 13.75 V tk = 150 sec. max. See note 24	If	2.9	3.3	A
4223	Tuner torque (room temp.)	-	starting dynamic at 2100 rpm See note 26	Torque Torque	--- ---	8 6	in-oz in-oz
4303	Heater cathode warm-up time	1	See note 5	tk	150	---	sec
4304	Pulse characteristics (1)	1	See notes 6 and 7	---	---	---	---
4306	Pulse voltage (1)	-	See notes 11 and 38	epy	20.0	23.0	kv
4250	Power output (1)	1	See notes 11 and 38	Po	200	270	W
4308	RF Bandwidth (1)	1	F1, F2 and F3 See notes 11 and 21	BW	---	2.0/tpc	MHz
4308	Minor lobe ratio (1)	1	F1, F2 and F3 See notes 11 and 21	Ratio	-10	---	db
4315	Stability (1)	1	F1, F2 and F3 See notes 11 and 18	MP	---	0.07	%
4223	Mechanical tuning range	1	Upper Limit Lower Limit	F F	9420 ---	--- 8680	MHz MHz
----	Tuning stability	-	F1 to F3 See note 36	Arcs	---	0.10	%
4303	Heater cathode warm-up time	2	See note 5	tk	150	---	sec
4304	Pulse characteristics (2)	2	See notes 6 and 7	---	---	---	---
4306	Pulse voltage (2)	2	See note 11	epy	19.0	23.0	kv
4308	RF Bandwidth (2)	2	F1, F2 and F3 See notes 11 and 21	BW	---	2.0/tpc	MHz
4308	Minor lobe ratio (2)	2	F1, F2 and F3 See notes 11 and 21	Ratio	-8	---	db
4315	Stability (2)	2	F1, F2 and F3 See notes 11 and 18	MP	---	0.07	%
4315	Starting stability	2	F2; See notes 18 and 30	MP	---	0.25	%
4250	Power output (2)	2	F1, F2 and F3 See notes 11 and 38	Po	140	---	watts

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 Note 1						
----	Tuning linearity	1	See note 40	ΔF	---	±25	MHz
----	Frequency drift rate	1	F1, F2 and F3 See note 19	ΔF/Δt	---	+150	kHz/s
4223	Tuner torque	2	Tbody = -55°C nonoperating	Dynamic	---	6	oz-in
			Tbody = 150°C max, operating See note 26	Static	---	8	oz-in
4223	Resettability	1	F3 -100 ±5 MHz F1 ±5 MHz See note 33	ΔF	---	1.2	MHz
	Quality conformance inspection, part 3						
4551	Life test provisions	-	Group D or S; VSWR = 1.5 min. cycled through λg every 15 minutes approximately See note 37	---	1500	---	cycles
				---	1250	---	hours
	Life Test End Points		See note 12				
4250	Power output	1	See note 11	Po	160	---	watts
4250	Power output	2	See note 11	Po	115	---	watts
4223	Mechanical tuning range	1	Upper Limit	F	9410	---	MHz
			Lower Limit	F	---	8690	MHz
4308	RF Bandwidth	1	F1, F2 and F3 See notes 11 and 21	BW	---	2.5/tp	MHz
4308	RF Bandwidth	2	F1, F2 and F3 See notes 11 and 21	BW	---	2.5/tp	MHz
4308	Minor lobes (1)	1	F1, F2 and F3 See note 11 and 21	Ratio	-7	---	db
4315	Stability (1)	1	See notes 11 and 18	MP	----	0.10	%
4315	Starting stability	1	F = F2 See notes 18 and 30	MP	---	.10	%
4315	Stability	1	F = F2 See notes 11 and 18	MP	----	.07	%
4306	Pulse voltage	1	F = F2 See note 11	epy	20	23	kv

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 (Continued)						
	Periodic-check tests		See note 8				
4027	Temperature coefficient	1	Tbody = 0°C to 150°C F = F1, F2 and F3 See notes 2 and 41	$\frac{\Delta F}{\Delta T}$	---	0.20	MHz/°C
1042	Shock	-	30G; 11 ±1 ms duration; no voltages See notes 13 and 33	---	---	---	----
----	Input capacitance	-	Cathode terminal to mounting plate	C	9	13	pF
1143	Air cooling	1	See notes 2 and 10	ΔT	---	79	°C
1032	Variable frequency vibration	1	F = F3 See note 14	----	---	---	----
4315	Stability (Curve A)	1	See note 15	MP	---	0.5	%
4318	Frequency modulation (Curve C)	1	See note 31	FM	---	1.0	MHz
4315	Stability (Curve C)	1	See note 15	MP	---	0.1	%
4223	Mechanical tuning fatigue	-	No voltages See note 16	----	10^5	---	cycles
4311	Pushing factor	1	I_b = 19 to 27 mA See notes 11 and 20	ΔF/Δib	---	100	kHz/a
4310	Pulling factor	1	See notes 11 and 34	ΔF	---	5	MHz
4308	Spectrum measurements	1	I_b = 19 and 27 mA F = F1, F2 and F3 See notes 11 and 20				
1031	Variable frequency vibration (nonoperating)	-	See notes 8 and 33	---	---	---	----
4302A	Low temperature frequency change	1	F = F1, F2 and F3 t_3 = 20 min. See note 42	ΔF	---	±10	MHz
4302A	High temperature frequency change	1	F = F1, F2 and F3 See note 43	ΔF	---	±10	MHz
1011	Humidity	-	See note 35	----	---	---	----
----	Acoustic noise	2	See note 9	MP FM	--- ---	0.1 ±0.5	% MHz

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NOTES:

1. The AQL for the combined defectives for attributes in Quality Conformance Inspection Part 1, excluding inoperatives and mechanical shall be 1 per cent. A tube having one or more defects shall be counted as one defective. Standard MIL-STD-105, Inspection Level II shall apply. For Part 2 the AQL percent defects shall be 6.5 percent, Inspection Level L6 shall apply.

   Rejection and resubmittals shall be in accordance with MIL-STD-105, Section 12.

2. The temperature shall be measured at the point indicated on Figure 1.

3. The tuning mechanism shall be capable of withstanding a static torque of 200 in-oz at the ends of its travel.

4. The magnetron shall be capable of normal operation without electrical breakdown with the input bushing in air at 11.0 pounds absolute.

5. Prior to the application of high voltage for test the cathode shall be allowed to warm-up to nominal operating temperatures. The tube under test shall be capable of operation under Test 1 or Test 2 conditions after 150 sec (max.) warm-up time. Upon application of anode voltage, the heater voltage shall be reduced according to the following:

	Du	Ib (mAdc)	If(A)	Ef(V) ± 5%
Standby	------	-----	3.1 (nom.)	13.75
Operate Test 1	0.001	27.5	0.1 (nom.)	0.2
Operate Test 2	0.0008	19.2	1.5 (nom.)	5.0

   For operation at power input levels differing from test conditions specified herein, the heater voltage should be adjusted according to the following formula:

   For Pi equal to or greater than 600 watts; $E_f = 0$

   For Pi between 375 and 600 watts; $E_f = 19.0 \times (1\text{-}Pi/600)^{+1.4}_{-0.7}$ volts

   For Pi less than 375 watts $E_f = 13.75\ (1\text{-}Pi/800)^{+1.4}_{-0.7}$ volts

6. The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of rise of the voltage pulse, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. See Method 4304, Figure 4304-2.

7. The rate of rise of voltage (rrv) shall be measured in accordance with Method 4304, except that the steepest tangent to the leading edge of the voltage pulse shall be measured above the 70 per cent amplitude point. Any capacitance used in the viewing circuit shall not exceed 6 picofarads (pf).

8. The magnetron shall be mounted in a resonance-free jig and vibrated with sinusoidal excitation in each of three mutually perpendicular planes through the following conditions: Curve B Figure 2; 3 hours.

9. The magnetron shall be capable of specified performance of Quality Conformance Inspection Part 1 and Part 2 while being subjected to acoustican energy level of 135 ±5 db with the spectrum shape as defined by MIL-STD-810, Method 515, t = 5 hours.

10. With a total airflow of approximately 3.0 lb/min. at approximately 760 mm. Hg., 25°C, divided equally and directed through the cooling fins toward the body of the tube at a maximum back pressure of 2.0 inch of water, the specified rise above ambient temperature shall not be exceeded. The test frequency may be the frequency of minimum power output between 8700 and 9400 MHz.

11. Though the requirements exist continually from F1 to F3, tests shall be performed F1, F2 and F3.

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NOTES: -Cont'd

12. If during life test, the tube does not meet the specified limits, it shall be recycled for an additional five cycles. At such time, the tests shall be repeated. The tube will be considered satisfactory if it passes the second test.

13. The tube shall be subjected to five shocks of the peak amplitude and duration specified herein in each of three mutually perpendicular directions in accordance with MIL-STD-202, Method 213, test condition G.

14. The magnetron shall be mounted in a resonance free jig and vibrated with sinusoidal excitation in each of three mutually perpendicular planes through the following conditions:

Curve A	Figure 2; 1.5 hours
Curve C	Figure 2; 3 hours

15. A missing pulse is defined as one whose energy with ± 1 per cent frequency range of the normal operating frequency is 70 per cent or less than that of a normal pulse. The tube shall be considered stable if the missing pulse limit specified herein is not exceeded during the vibration time and acceleration levels as specified in the operating vibration test.

16. The tuner shaft shall be continuously driven at a speed of 2100 ± 100 RPM. At the completion of the test the tube shall meet Test condition 1 requirements. The backlash shall not exceed 1.5 MHz. During the test, the tuning shaft shall not be lubricated. A cycle consists of two complete excursions each in opposite directions through the tuning range of the magnetron.

17. With input and output properly pressurized, the magnetron shall pass the stability requirements of Note 18.

18. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP) due to any causes, are considered to be missing if the R. F. energy is less than 70 per cent of the normal energy level. The stability shall be measured when a VSWR of 1.5 minimum is introduced in the load at a distance of 1 ± 0.05 meters from the magnetron flange and the phase is adjusted to produce maximum instability. The missing pulse test shall be counted during any three consecutive minute interval of a six-minute test period.

19. With cooling air supplied to the tube in such a manner that it can be controlled in flow from zero to maximum, adjust and record the flow for a body temperature of 80°C ±5°C. After thermal equillibrium has been reached, record magnetron frequency. Remove cooling air completely in less than one second, and record frequency and anode temperature at the end of the first 15 seconds. The rate of frequency drift shall not exceed the limit specified. Also, with body temperature stabilized at 145°C ± 5°C and frequency recorded, suddenly (1 sec. max.) adjust air flow to that recorded for Tbody = 80 ± 5°C, and measure frequency and Tbody at end of first 15 seconds. The rate of frequency drift shall not exceed the limit specified. Conditions are Oscillation (1) and frequencies are F1, F2 and F3.

20. The Pushing factor shall be measured in two steps, 4 amperes each (19A to 23A, 23A to 27A) and no value shall exceed the limits specified herein. The peak current through the magnetron shall alternately be the limits as specified under this test condition. These tests shall be run to exclude the effects of thermal drift and frequency instability not due to pushing.

21 The R. F. Bandwidth and Minor Lobes shall be within the limits specified herein when a VSWR of 1.5 minimum is introduced in the load at a distance of 1 ± 0.05 meters from the magnetron flange and the phase is adjusted at the start of each measurement to produce maximum degradation. A satisfactory spectrum is one whose slope does not change sign more than once for power levels greater than the specified db below its peak.

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NOTES: -Cont'd

22. With the magnetron mounted on the test fixture and with the specified air pressure applied so as to surround the entire input terminal beyone the magnetron mounting plate, the entire magnetron and fixture are to be submerged in water. No bubbles are allowed in a one-minute interval.

23. With the magnetron output flange clamped against a modified choke flange (UG-52/B/U) and using a gasket per MS-90064-14, and with the specified air pressure applied to the interior of the waveguide, submerge entire magnetron and fixture in water. No bubbles are allowed in a one-minute interval.

24. The magnetron shall be capable of withstanding a heater surge current of 3.5 times the rated heater current. (See Figure 3)

25. The minimum air pressure to assure prevention of electrical breakdown in the output coupling shall be 15 psia for voltage standing wave ratios up to 1.5 with phase shift variable over 360 electrical degrees. The magnetron shall be coupled directly to UG-52/B/U choke flange in accordance with MIL-F-3922/53 modified so that mounting holes provide clearance for #8 bolts. For VSWR of 1:1, the pressure is 11 psia.

26. After tuner is driven into stop, it shall require no more than 20 in-oz of torque to reverse the tuner direction. The reverse torque shall not be an impulse, but rather a gradual application. The tuner shall have been driven into the stops by means of a servo control motor (Westinghouse P-Number 579R936H01) driven at full excitation, 40 Vrms, through a gear reduction of 4.28:1 to the magnetron tuner shaft. The total drive system inertia referred to the magnetron tuner is 250 gm-cm.

27. The frequency obtained by turning the tuner to a given dial setting in one direction shall be reproducible within the limits specified herein when returning to that same setting from the opposite direction after thermal equilibrium. If hysteresis is present, it shall be measured in the following way: Set the tube frequency to 250 MHz above center band and note the shaft angle. Rotate the shaft angle +90 degrees from the initial angle, then -90 degrees, which returns the shaft to its initial position. Value of difference frequency shall not exceed limit specified. Repeat to F1.

28. Unless otherwise specified, all tests required by this specification shall be made under the following atmospheric conditions:

Temperature	- 25 ± 10°C
Relative Humidity	- 90 per cent or less
Barometric Pressure	- local standard

29. The power output shall be continuous over the range from 8700 to 9400 MHz. At no frequency in this range shall the power output be less than the minimum value specified herein.

30. After the holding period, the anode voltage shall be applied 150 seconds (maximum) after the application of the standby heater voltage. The missing pulse count test interval of 3 minutes shall start immediately after the application of anode voltage.

31. For frequency modulation measurements, the maximum peak to peak frequency deviation shall not exceed the amount specified herein. The test shall be run excluding Thermal Drift and Pushing effects. The test equipment shall have a bandpass of 10 MHz minimum at the 3 db points. Frequency modulation is defined as inter-pulse frequency changes.

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NOTES: -Cont'd

32. In addition to regular markings, the tuner dial settings for the following frequencies shall be marked on the tube during final testing by the manufacturer. The accuracy of these settings shall be ± 5 MHz at the start of life under conditions of Test Condition 1 with the body temperature approximately 800°C. measured at the point specified on Figure 1 when tuning is performed in the order of increasing the frequency,

FREQUENCY	DIAL SETTING	FREQUENCY	DIAL SETTING
8700 MHz	---	9100 MHz	---
8800	---	9200	---
8900	---	9300	---
9000	---	9400	---

33. At the completion of this test, the tube shall meet the power output and voltage requirements of Test 1 of Quality Conformance Inspection, Part 1, and Tuner Drive Torque requirements of Quality Conformance Inspection, Part 2.

34. The frequency deviation (maximum frequency minus minimum frequency) shall not exceed the limits specified herein when a VSWR of 1.5 minimum, introduced into the load at a distance 1 ± 0.05 meters from the magnetron flange, is varied throughout all phases.

35. The waveguide output flange and the high voltage input bushing shall be enclosed for the Humidity and Barometric tests.

36. The tube shall be tuned from F1 to F3 to F1 at 2100 ± 10% rpm. The three minute test interval shall begin within 10 seconds after application of high voltage.

37. The cycled life test shall be performed under conditions listed below. Only the operating time under Test 1 and Test 2 conditions is creditable for life-hour computation. Group S life test shall consist of Sequential Life Test and Initial Life Test only. Certified life test data accrued under Government contracts may be substituted for Initial Life Test providing such data meets the specified criteria for creditation.

Condition	ib (a)	If (A)	Duration
Standby	0	3.1 ± 0.2	2.5 minutes
Test 1	27.5	0.1 ± 0.1	25.0 minutes
Test 2	24.0	1.5 ± 0.6	25.0 minutes
Off	0	0	7.5 minutes

Magnetron frequency shall be changed 100 MHz at approximately every 50 hours, starting at low frequency.

38. The tube shall meet the limit specified herein after it has been subjected to the shock and vibration tests.

39. The frequency range F1 to F3 MHz shall be traversed by a tuning shaft rotation of 100 ± 10 turns. A mechanical over-run of 3.0 shaft turns (min.) shall be provided at each end of the tuning range before contacting the mechanical stops.

40. The tuning linearity curves of frequency versus dial settings shall be derived from not less than 8 approximately evenly spaced data points when tuning in the direction of increasing frequency across the band. This curve shall not deviate from a least squares fit straight line curve derived from the data points by more than ± 25 MHz. Conditions are Test 1: Tbody = 80 ± 10°C.

41. Temperature measurements shall be made only after thermal equilibrium has been reached. The frequency shall be measured at the extremes of any 30°C temperature range.

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42. The tube shall be stabilized non-operating at -54°C. At t = 0, filaments and 3 lbs/min. cooling air at 30°C are applied. At t = 150 sec anode voltage shall be applied. At t = 300 sec, the tubes shall not have a greater frequency shift than that specified. T_3 (at 20 minutes) shall be measured and the stabilized frequency change shall be no greater than the specified maximum.

43. The tube shall be stabilized non-operating at 95°C. At t = 0, filaments and 3 lbs/min. cooling air at 30°C are applied. At t = 150 sec, anode voltage shall be applied. At t = 300 sec, the tube shall not have a greater frequency shift than that specified. T_3 (at 20 minutes) shall be measured and the stabilized frequency change shall be no greater than the specified maximum.

44. The first article (preproduction) sample required for inspection and the acceptance criteria shall consist of four tubes. Life test shall subsequently be performed on three of the four sample tubes. First article testing shall include all tests and requirements of this specification and table 1 of MIL-E-1, where required. The four sample tubes shall be subjected to all electrical tests prior to undergoing environmental testing. The contractor/manufacturer shall provide test reports and written assurance that:

    a. The first article sample is representative of the product to be submitted under the contract.

    b. Production articles will be manufacturer from new materials and in the facility that produced the sample tubes.

    A test plan shall be submitted to the contracting officer for approval prior to proceeding with first article testing. The test plan shall list all production test and laboratory test facilities to be utilized in first article testing and production phases of the contract. Upon receipt of approval of the test plan, first article testing may proceed. When testing has been completed, three copies of the test report shall be prepared in accordance with MIL-STD-831, certified by a government representative and submitted with the test samples to the contracting officer for approval.

Custodian:

Air Force - 85

Review activities:
Air Force - 70, 80

User activities:
Air Force - 11

Preparing activity:

Air Force - 85

(Project 5960-F972)

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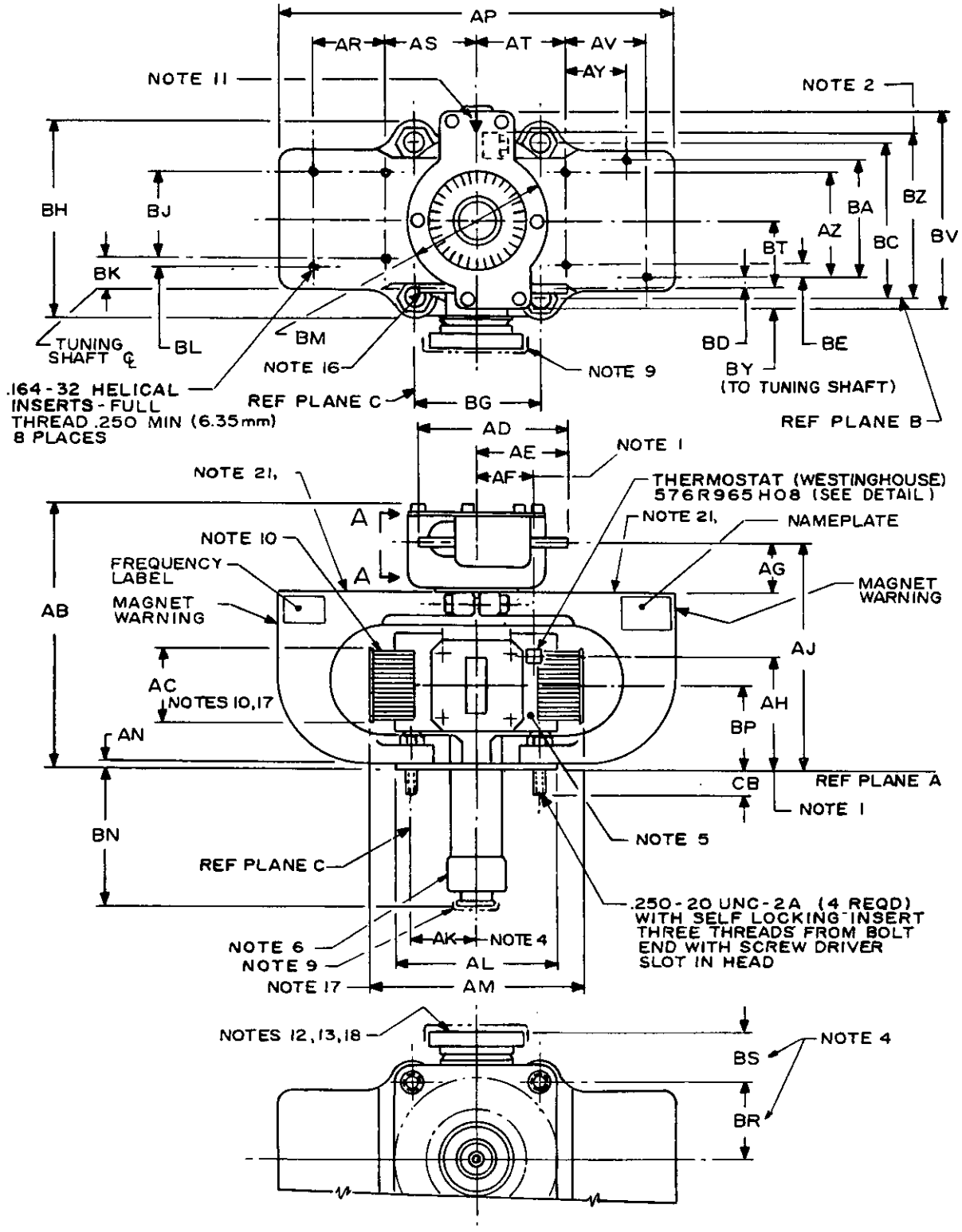
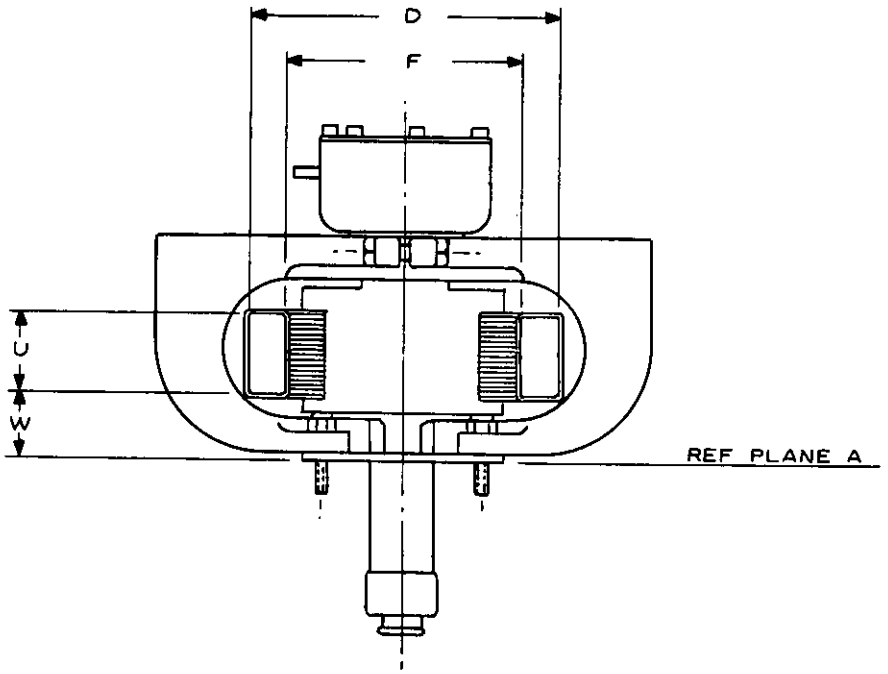
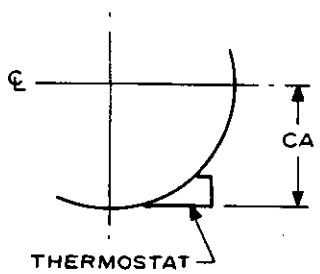


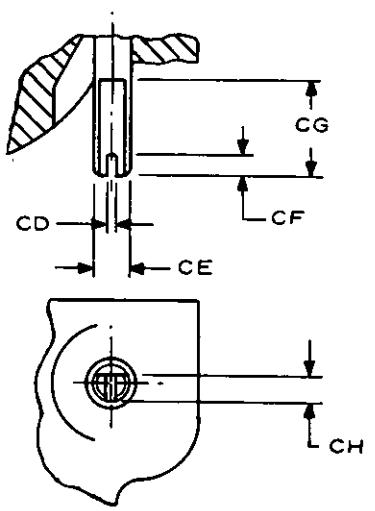
FIGURE 1 Outline drawing tube type 8945

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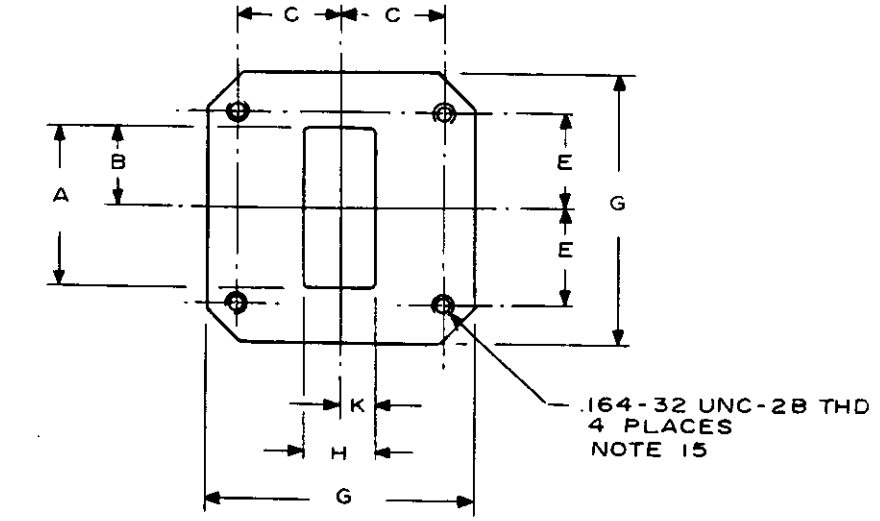
TOP VIEW
SHOWING DISTANCE THERMOSTAT
PROTRUDES FROM ℄ OF TUBE



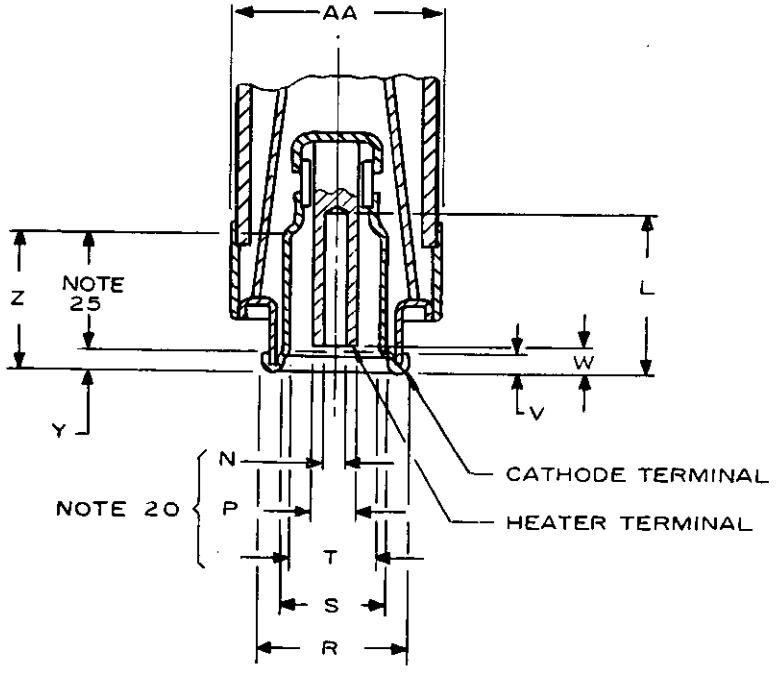
VIEW A A
TUNING SHAFT
(TYPICAL BOTH ENDS)

FIGURE 1. Outline drawing tube type 8945 (Continued)

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OUTPUT FLANGE
NOTE 21



INPUT CONNECTION
NOTE 19

FIGURE 1. Outline drawing tube type 8945 (Continued)

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PT.1				
A	1.119	1.125	28.42	28.58
B	.5595	.5625	14.21	14.29
C	.735	.739	18.67	18.77
E	.674	.678	17.12	17.22
H	.494	.500	12.55	12.70
K	.247	.250	6.27	6.35
L	.750	-	19.05	-
M	.125	.187	3.18	4.75
N	.164	.174	4.17	4.42
P	.234	.266	5.94	6.76
R	.825	.838	20.96	21.29
S	.603	.617	15.32	15.68
T	.532	.545	13.51	13.84
V	.115	.135	2.92	3.43
AE	1.390	1.410	35.31	35.81
AG	.965	.985	24.51	25.02
AJ	4.338	4.462	110.19	113.33
AK	1.204	1.296	30.58	32.92
AS	1.646	1.671	41.81	42.44
AT	1.840	1.865	46.74	47.37
AV	1.557	1.567	39.55	39.80
AY	1.307	1.317	33.20	33.45
BG	2.490	2.510	63.25	63.75
BN	2.606	2.702	66.19	68.63
BP	1.610	1.702	40.89	43.23
BS	.874	.938	22.20	23.83
BZ	-	3.125	-	79.38
CB	.375	.499	9.53	12.67
CD	.040	.045	1.02	1.14
CE	.187	.189	4.75	4.80

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2				
G	1.860	1.890	47.24	48.01
Y	-	.156	-	3.96
Z	.562	-	14.27	-
AA	-	1.500	-	38.10
AB	-	5.500	-	139.70
AD	2.578	2.598	65.48	65.99
AN	.031	-	.79	-
AP	7.500	7.686	190.50	195.22
AR	1.495	1.505	37.97	38.23
AZ	1.932	1.942	49.08	49.33
BA	2.307	2.317	58.60	58.85
BC	2.990	3.010	75.95	76.45
BD	.178	.198	4.52	5.03
BE	.370	.380	9.40	9.65
BJ	1.526	1.536	38.76	39.01
BK	.428	.448	10.87	11.38
BL	.213	.223	5.41	5.66
BM	2.625	2.749	66.68	69.82
BR	1.454	1.546	36.93	39.27
BT	1.240	1.260	31.50	32.00
BY	.425	.455	10.80	11.56
CA	1.800		45.72	
CF	.120	.130	3.05	3.30
CG	.615	.635	15.62	16.13
CH	.115	.195	2.92	4.95
NOMINAL				
AC	1.200		30.48	
AF	1.220		30.99	
AH	2.300		58.42	
AL	3.406		86.51	
AM	5.400		137.16	
BH	3.750		95.25	
BV	4.000		101.60	
D	5.600		142.24	
F	3.915		99.44	
U	.920		23.37	
W	1.165		29.59	

FIGURE 1. Outline drawing tube type 8945 (Continued)

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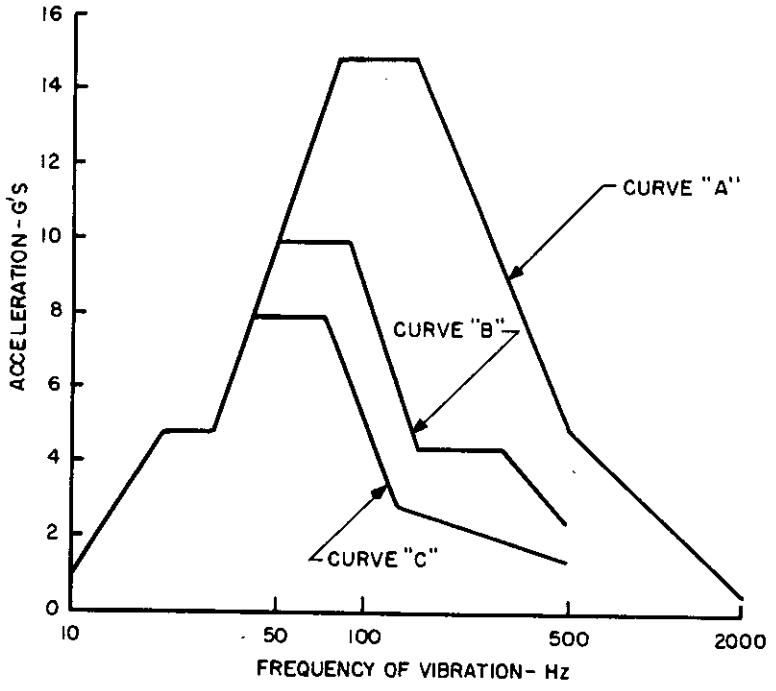


FIGURE 2. Input vibration requirements

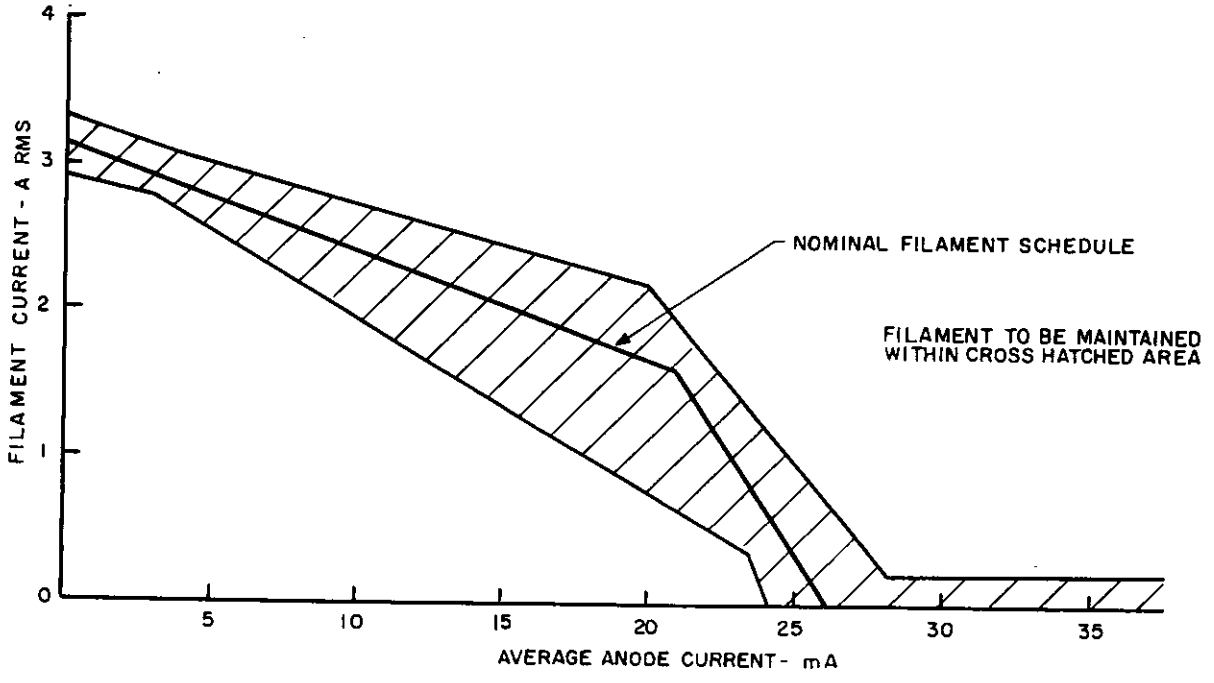


FIGURE 3. Filament current schedule

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NOTES:

1. This dimension applies to the location of the thermostatic switch.
2. This dimension applies to the magnet clamp attachment only.
3. This dimension applies to Diameter T.
4. Includes angular, as well as lateral, deviations.
5. Anode temperature to be measured at this point. The magnetron serviceability will not be impaired with a screw inserted, or removed, from this threaded hole.
6. Input bushing temperature to be measured at this point.
7. For Vibration and Shock testing, the planes of testing shall be reference planes "A", "B", "C".
8. WARNING: Maintain minimum clearance 4 inches between magnet and magnetic materials (magnets, steel tools, plates, etc.). Use no magnetic inspection tools.
9. Protective closure. To be removed before magnetron is used, and attached when the tube is not in use.
10. The inner laminations of the cooling fins are not painted. Overspray of protective paint is permissible.
11. Indicates direction of body cooling air flow.
12. Mates with choke flange, UG-52 B/U modified to provide clearance instead of threaded holes.
13. This face has a satin type (smooth, dull) finish of approximately 63 micro-inches.
14. This identifies a North seeking pole. This may be tested by observing which compass needle points to the geographic South. The same compass needle will point toward the magnetron North seeking pole when it is brought near the input stem of the magnetron. Care should be taken to insure that the magnetron field does not reverse the polarity of the compass.
15. A plane passing through the axis of two threaded holes perpendicular to the face of the output flange must be parallel to planes "A" and "C" within .030.
16. The frequency increases as this end of the tuner input shaft is driven in a counter-clockwise direction.
17. This dimension refers to radiator fin size only.
18. The face of the waveguide flange shall be flat within .010 total indicator reading. The surface of this flange shall be of such quality that a hermetic type seal can be effected.
19. Input connection mates with Jettron Products Inc., Hanover, N.J.. (Connectors Cat. #90-006 and 90-030, or equivalent).
20. Heater terminal and cathode terminal shall be concentric within .010.
21. These surfaces to bear removable protective covers. Surfaces to be finished 125 microinch (min) Chromate conversion coating per MIL-C-5541, Class 3, color 30257 per FED-STD-595.

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